

MATH 3330 Homework 4

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Question 1

Let $X = \{a, b, c\}$. How many topologies are on X ?

The following are unique topologies on X :

1. $\{\emptyset, X\}$
2. $\{\emptyset, \{a\}, X\}$
3. $\{\emptyset, \{b\}, X\}$
4. $\{\emptyset, \{c\}, X\}$
5. $\{\emptyset, \{a, b\}, X\}$
6. $\{\emptyset, \{a, c\}, X\}$
7. $\{\emptyset, \{b, c\}, X\}$
8. $\{\emptyset, \{a\}, \{a, b\}, X\}$
9. $\{\emptyset, \{a\}, \{a, c\}, X\}$
10. $\{\emptyset, \{a\}, \{b, c\}, X\}$
11. $\{\emptyset, \{b\}, \{a, b\}, X\}$
12. $\{\emptyset, \{b\}, \{a, c\}, X\}$
13. $\{\emptyset, \{b\}, \{b, c\}, X\}$
14. $\{\emptyset, \{c\}, \{a, b\}, X\}$
15. $\{\emptyset, \{c\}, \{a, c\}, X\}$
16. $\{\emptyset, \{c\}, \{b, c\}, X\}$
17. $\{\emptyset, \{a\}, \{b\}, \{a, b\}, X\}$
18. $\{\emptyset, \{a\}, \{c\}, \{a, c\}, X\}$
19. $\{\emptyset, \{b\}, \{c\}, \{b, c\}, X\}$
20. $\{\emptyset, \{a\}, \{a, b\}, \{a, c\}, X\}$
21. $\{\emptyset, \{b\}, \{a, b\}, \{b, c\}, X\}$
22. $\{\emptyset, \{c\}, \{a, c\}, \{b, c\}, X\}$
23. $\{\emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, X\}$
24. $\{\emptyset, \{a\}, \{b\}, \{a, b\}, \{b, c\}, X\}$
25. $\{\emptyset, \{a\}, \{c\}, \{a, b\}, \{a, c\}, X\}$
26. $\{\emptyset, \{a\}, \{c\}, \{a, c\}, \{b, c\}, X\}$
27. $\{\emptyset, \{b\}, \{c\}, \{a, b\}, \{b, c\}, X\}$

28. $\{\emptyset, \{b\}, \{c\}, \{a, c\}, \{b, c\}, X\}$

29. $\{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, X\}$

There are a total of 29 unique topologies on X .

Question 2

Let $\mathcal{T}$ consist of $\mathbb{R}, \emptyset$, and all infinite open intervals in the form of (a, ∞) , where a is real. Show that $\mathcal{T}$ is a topology. What if we restrict $\mathcal{T}$ to $\mathbb{Q}$?

Proof: Let $\mathcal{T}$ consist of $\mathbb{R}, \emptyset$, and all infinite open intervals in the form of (a, ∞) , where a is real. Let $\mathcal{U} \subseteq \mathcal{T}$ a subcollection of $\mathcal{T}$. In order for $\mathcal{T}$ to be a topology, it must satisfy all of the following conditions:

1. If $\mathcal{T}$ is a topology on a set X , then $X, \emptyset \in \mathcal{T}$.
2. The union of elements for any subcollection of $\mathcal{T}$ is in $\mathcal{T}$.
3. The intersection of elements of any finite subcollection of $\mathcal{T}$ is in $\mathcal{T}$.

1. By the definition of $\mathcal{T}$, since $X = \mathbb{R}$, $X, \emptyset \in \mathcal{T}$.

2. If $\mathbb{R} \in \mathcal{U}$, then $\bigcup \mathcal{U} = \mathbb{R} \in \mathcal{T}$. If $\mathcal{U}$ contains only $\emptyset$, then trivially $\bigcup \mathcal{U} = \emptyset \in \mathcal{T}$. Now consider the case when $\mathbb{R} \notin \mathcal{U}$ and $\mathcal{U}$ contains collections other than $\emptyset$ ($\mathcal{U}$ contains at least one nonempty set in the form (a, ∞) for $a \in \mathbb{R}$). Let $A = \{a \in \mathbb{R} \mid (a, \infty) \in \mathcal{U}\}$. Since $\mathcal{U}$ contains at least one nonempty set, $A \neq \emptyset$. We will show that $\bigcup_{(a, \infty) \in \mathcal{U}} (a, \infty) = (\inf A, \infty)$ by double inclusion:

- $\subseteq$: If $x \in \bigcup_{(a, \infty) \in \mathcal{U}} (a, \infty)$, then $x > a$ for some $a \in A$, and since $\inf A \leq a$, we get $\inf A \leq a < x$. So, $x > \inf A$ by transitivity, so $x \in (\inf A, \infty)$.
- $\supseteq$: If $x \in (\inf A, \infty)$, then $\inf A < x$. By the definition of the infimum, there must exist some $a \in A$ such that $\inf A \leq a < x$, so $x \in \bigcup_{(a, \infty) \in \mathcal{U}} (a, \infty)$.

Therefore, $\bigcup_{(a, \infty) \in \mathcal{U}} (a, \infty) = (\inf A, \infty)$, so the union of elements from any subcollection $\mathcal{U}$ of $\mathcal{T}$ is in $\mathcal{T}$.

3. If $\emptyset \in \mathcal{U}$, then trivially $\bigcap \mathcal{U} = \emptyset \in \mathcal{T}$. Now consider $(a_1, \infty), \dots, (a_n, \infty)$ for $n \in \mathbb{N}$ and $a_1, \dots, a_n \in \mathbb{R}$. A point x is in the intersection of $(a_1, \infty), \dots, (a_n, \infty)$ if $x > a_1, \dots, x > a_n$. So, $x > \max\{a_1, \dots, a_n\}$, and thus $\bigcap_{i=1}^n (a_i, \infty) = (\max\{a_1, \dots, a_n\}, \infty)$. Since $\max\{a_1, \dots, a_n\} \in \mathbb{R}$, $(\max\{a_1, \dots, a_n\}, \infty) \in \mathcal{T}$, so the intersection of elements of any finite subcollection of $\mathcal{T}$ is in $\mathcal{T}$.

Therefore, $\mathcal{T}$ is a topology.

Restricting to $\mathbb{Q}$: We will define $\mathcal{T}_{\mathbb{Q}}$ as $\{U \cap \mathbb{Q} \mid U \in \mathcal{T}\}$. Then $\mathcal{T}_{\mathbb{Q}}$ becomes:

$$\mathcal{T}_{\mathbb{Q}} = \{\emptyset, \mathbb{Q}\} \cup \{(a, \infty) \cap \mathbb{Q} \mid a \in \mathbb{R}\}$$

1. $\emptyset = \emptyset \cap \mathbb{Q} \in \mathcal{T}_{\mathbb{Q}}$ and $\mathbb{Q} = \mathbb{R} \cap \mathbb{Q} \in \mathcal{T}_{\mathbb{Q}}$.

2. Take any subcollection $U_n \cap \mathbb{Q}_{n \in \mathbb{N}} \subseteq \mathcal{T}_{\mathbb{Q}}$ with each $U_n \in \mathcal{T}$. Then

$\bigcup_{n \in \mathbb{N}} (U_n \cap \mathbb{Q}) = (\bigcup_{n \in \mathbb{N}} U_n) \cap \mathbb{Q}$. Since $\bigcup_{n \in \mathbb{N}} U_n \in \mathcal{T}$ (because $\mathcal{T}$ is a topology on $\mathbb{R}$), it follows that $(\bigcup_{n \in \mathbb{N}} U_n) \cap \mathbb{Q} \in \mathcal{T}_{\mathbb{Q}}$. Therefore $\mathcal{T}_{\mathbb{Q}}$ is closed under arbitrary unions.

3. Let $U_1, \dots, U_n \in \mathcal{T}$ for $n \in \mathbb{N}$. Then $\bigcap_{i=1}^n (U_i \cap \mathbb{Q}) = (\bigcap_{i=1}^n U_i) \cap \mathbb{Q}$. Since $\bigcap_{i=1}^n U_i \in \mathcal{T}$ (because $\mathcal{T}$ is closed under finite intersections), it follows that $(\bigcap_{i=1}^n U_i) \cap \mathbb{Q} \in \mathcal{T}_{\mathbb{Q}}$. Thus $\mathcal{T}_{\mathbb{Q}}$ is closed under finite intersections.

Therefore $\mathcal{T}_{\mathbb{Q}}$ is also a topology.

Question 3

Let X be a topological space and A be a subset of X . Suppose that for each $x \in A$ there is an open set U containing x such that $U \subseteq A$. Show that A is open in X .

Proof: Let X be a topological space and $A \subseteq X$. Suppose that for each $x \in A$ there is an arbitrary open set U_x containing x such that $U_x \subseteq A$. We must show that A is open in X . Consider the union $U = \bigcup_{x \in A} U_x$. U must be open since the union of open sets is open. We will show that $U = A$.

First, if $y \in U$, then $y \in U_x$ for some $x \in A$, and since $U_x \subseteq A$, $y \in A$, so $U \subseteq A$.

Then, if $y \in A$, then by assumption $y \in U_y$, where for $y \in A$, $U_y \subseteq U$ contains y . Then, $y \in U$. Thus $A \subseteq U$.

Therefore, $A = U$, and since U is open in X , A must also be open in X .

Question 4

Show that if $\mathcal{A}$ is a basis for a topology on X , then the topology generated by $\mathcal{A}$ equals the intersection of all topologies on X that contain $\mathcal{A}$. Prove the same if $\mathcal{A}$ is a subbasis.

Proof: Let $\mathcal{A}$ be a basis for a topology on X and let $\mathcal{T}_{\mathcal{A}}$ be the topology generated by $\mathcal{A}$, so $\mathcal{T}_{\mathcal{A}} = \bigcup_{U \subseteq \mathcal{A}} U$. Also, let $\mathcal{T}$ be the intersection of all topologies on X that contain $\mathcal{A}$. We must show that $\mathcal{T}_{\mathcal{A}} = \mathcal{T}$ by double inclusion.

$\subseteq$: Let $U \in \mathcal{T}_{\mathcal{A}}$. Then, $U = \bigcup \mathcal{U}$ for $\mathcal{U} \subseteq \mathcal{A}$ since $U \in \mathcal{T}_{\mathcal{A}}$ and $\mathcal{T}_{\mathcal{A}}$ is the topology generated by the basis $\mathcal{A}$. Now let τ be some arbitrary topology on X . Then $\mathcal{U} \subseteq \mathcal{A} \subseteq \tau$. Since τ is a

topology, it is closed under arbitrary unions, so $\bigcup \mathcal{U} \in \tau$. Thus $U \in \tau$ for each topology τ , since $U = \bigcup \mathcal{U}$. Therefore, U lies in the intersection of all τ , so $\mathcal{T}_\mathcal{A} \subseteq \mathcal{T}$.

$\supseteq$: $\mathcal{T}_\mathcal{A}$ is a topology on X that contains $\mathcal{A}$ since it was constructed from the basis $\mathcal{A}$. Then, $\mathcal{T}_\mathcal{A}$ is one of the topologies intersected to form $\mathcal{T}$. Since an intersection of a family of sets is always contained within each member of the family, $\mathcal{T} \subseteq \mathcal{T}_\mathcal{A}$.

Thus $\mathcal{T}_\mathcal{A} = \mathcal{T}$ and therefore the topology generated by $\mathcal{A}$ equals the intersection of all topologies on X that contain $\mathcal{A}$.

If $\mathcal{A}$ is a subbasis: Now, let $\mathcal{A}$ be a subbasis for a topology on X , and let $\mathcal{T}_\mathcal{A}$ be the topology generated by $\mathcal{A}$. Let $\mathcal{B} = \{\bigcap A_i \mid A_i \in \mathcal{A}\}$ be the collection of all finite intersections of elements of $\mathcal{A}$. Then $\mathcal{B}$ is a basis for $\mathcal{T}_\mathcal{A}$, so every element of $\mathcal{T}_\mathcal{A}$ can be written as a union of elements of $\mathcal{B}$, that is, $\mathcal{T}_\mathcal{A} = \{\bigcup \mathcal{U} \mid \mathcal{U} \subseteq \mathcal{B}\}$. Also, let $\mathcal{T}$ be the intersection of all topologies on X that contain $\mathcal{A}$. We must show that $\mathcal{T}_\mathcal{A} = \mathcal{T}$ by double inclusion.

$\subseteq$: Let $U \in \mathcal{T}_\mathcal{A}$. Then $U = \bigcup \mathcal{U}$ for some $\mathcal{U} \subseteq \mathcal{B}$, since $\mathcal{B}$ is a basis for $\mathcal{T}_\mathcal{A}$. Now let τ be some arbitrary topology on X such that $\mathcal{A} \subseteq \tau$. Since τ is a topology, it is closed under finite intersections, so every element of $\mathcal{B}$ lies in τ . So $\mathcal{U} \subseteq \mathcal{B} \subseteq \tau$. Since τ is also closed under arbitrary unions, we have $\bigcup \mathcal{U} \in \tau$. Thus $U \in \tau$ for each topology τ containing $\mathcal{A}$. Therefore U lies in the intersection of all such τ , so $\mathcal{T}_\mathcal{A} \subseteq \mathcal{T}$.

$\supseteq$: $\mathcal{T}_\mathcal{A}$ is a topology on X that contains $\mathcal{A}$, since $\mathcal{A} \subseteq \mathcal{B} \subseteq \mathcal{T}_\mathcal{A}$ because each $A \in \mathcal{A}$ is a finite intersection with $n = 1$. Therefore $\mathcal{T}_\mathcal{A}$ is one of the topologies intersected to form $\mathcal{T}$. Since an intersection of a family of sets is always contained within each member of the family, $\mathcal{T} \subseteq \mathcal{T}_\mathcal{A}$.

Thus $\mathcal{T}_\mathcal{A} = \mathcal{T}$, and therefore the topology generated by the subbasis $\mathcal{A}$ equals the intersection of all topologies on X that contain $\mathcal{A}$.

Question 5

A map $f : X \rightarrow Y$ is said to be an open map if for every open set U of X , the set $f(U)$ is open in Y . Show that $\pi : X \times Y \rightarrow Y$ is an open map.

Proof: Let $f : X \rightarrow Y$ be an open map. That is, for every open set U of X , the set $f(U)$ is open in Y . Let $V \subseteq X \times Y$ be open. We must show that for $\pi : X \times Y \rightarrow Y$, $\pi(V)$ is also open in Y .

Then, for all open subsets $A \subseteq X$ and $B \subseteq Y$, we have

$$V = \bigcup A \times B$$

Since $\pi(A \times B) = B$ for all subsets A and B , and the projection distributes over unions, then for all subsets A and B , we get

$$\pi(V) = \pi\left(\bigcup A \times B\right) = \bigcup \pi(A \times B) = \bigcup B$$

Since B is open in Y , so is $\pi(V)$ since it is the union of open sets, so $\pi : X \times Y \rightarrow Y$ is an open map.

Question 6

Show that the countable collection

$$\{(a, b) \times (c, d) \mid a < b, c < d, a, b, c, d \in \mathbb{Q}\}$$

is a basis for $\mathbb{R}^2$ with the standard topology.

Proof: In order for the collection $X = \{(a, b) \times (c, d) \mid a < b, c < d, a, b, c, d \in \mathbb{Q}\}$ to be a basis $\mathcal{B}$ for $\mathbb{R}^2$ with the standard topology, the following must be true:

1. For each $x \in X$, there is at least one basis element in $\mathcal{B}$ containing it.
2. If $B_1, B_2 \in \mathcal{B}$ and $x \in B_1 \cap B_2$, then $x \in B_3 \subseteq B_1 \cap B_2$.

1. Let $(x, y) \in \mathbb{R}^2$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, we can choose rationals a, b, c, d such that $a < x < b$ and $c < y < d$. So, $(x, y) \in (a, b) \times (c, d)$, and thus is in $\mathcal{B}$, so each element in the collection has at least one basis element containing it.

2. Let $B_1 = (a_1, b_1) \times (c_1, d_1)$ and $B_2 = (a_2, b_2) \times (c_2, d_2)$ be elements of $\mathcal{B}$, and suppose that $(x, y) \in B_1 \cap B_2$. We must show that there exists some $B_3 \in \mathcal{B}$ such that $(x, y) \in B_3 \subseteq B_1 \cap B_2$.

$$B_1 \cap B_2 = (\max(a_1, a_2), \min(b_1, b_2)) \times (\max(c_1, c_2), \min(d_1, d_2))$$

Since $(x, y) \in B_1 \cap B_2$, then $\max(a_1, a_2) < x < \min(b_1, b_2)$ and $\max(c_1, c_2) < y < \min(d_1, d_2)$. Because $\mathbb{Q}$ is dense, we want to choose $a', b', c', d' \in \mathbb{Q}$ such that

$$\begin{aligned} \max(a_1, a_2) &< a' < x < b' < \min(b_1, b_2) \\ \max(c_1, c_2) &< c' < y < d' < \min(d_1, d_2) \end{aligned}$$

Then, by our construction, $B_3 = (a', b') \times (c', d') \in \mathcal{B}$ and $(x, y) \in B_3 \subseteq B_1 \cap B_2$.

Therefore, $\mathcal{B}$ is a basis for $\mathbb{R}^2$ with the standard topology.

Question 7

If L is a straight line in the plane, describe the topology L inherits as a subspace of $\mathbb{R}_l \times \mathbb{R}$ and as a subspace of $\mathbb{R}_l \times \mathbb{R}_l$.

Note: there are three different cases for a line L :

1. Vertical: $L = \{x\} \times \mathbb{R}$
2. Horizontal : $L = \mathbb{R} \times \{y\}$
3. Sloped: $L = \{(x, mx + b) \mid x, m, b \in \mathbb{R}, m \neq 0\}$

For $\mathbb{R}_l \times \mathbb{R}$:

The basis for $\mathbb{R}_l \times \mathbb{R}$ consists of sets in the form $[a, b) \times (c, d)$.

- Vertical Line: An open set intersects with L in $\{x\} \times (c, d)$ only when $a \leq x < b$. This gives the standard topology.
- Horizontal Line: An open set intersects with L in $[a, b) \times \{y\}$ when $c < y < d$. This gives the lower limit topology $\mathbb{R}_l$.
- Sloped Line: Parametrize L by $x \mapsto (x, mx + b)$. A basis element $[a, b) \times (c, d)$ intersects L in the set of x with $x \in [a, b)$ and $mx + b \in (c, d)$, i.e. $x \in (\frac{c-b}{m}, \frac{d-b}{m})$ (reversing the inequality if $m < 0$). Thus the intersection corresponds to $x \in [a, b) \cap (\frac{c-b}{m}, \frac{d-b}{m})$. If $m > 0$, such intersections generate half-open intervals $[\alpha, \beta)$, giving the lower limit topology $\mathbb{R}_l$. If $m < 0$, the interval from the y -constraint is reversed, and the intersections are open intervals (α, β) , giving the standard topology.

For $\mathbb{R}_l \times \mathbb{R}_l$:

The basis for $\mathbb{R}_l \times \mathbb{R}_l$ consists of sets in the form $[a, b) \times [c, d)$.

- Vertical Line: An open set intersects L in $\{x\} \times [c, d)$ only when $a \leq x < b$. This gives the *lower limit topology* $\mathbb{R}_l$.
- Horizontal Line: An open set intersects L in $[a, b) \times \{y\}$ when $c \leq y < d$. This gives the *lower limit topology* $\mathbb{R}_l$.
- Sloped Line: Parametrize L by $x \mapsto (x, mx + b)$. A basis element $[a, b) \times [c, d)$ intersects L in the set of x with $x \in [a, b)$ and $x \in [\frac{c-b}{m}, \frac{d-b}{m})$ when $m > 0$. This intersection of two half-open intervals of the same type is again of the form $[\alpha, \beta)$, giving the *lower limit topology* $\mathbb{R}_l$. If $m < 0$, the constraint from y reverses to $x \in (\frac{d-b}{m}, \frac{c-b}{m}]$, so intersections with $[a, b)$ yield intervals of the form $(\alpha, \beta]$, giving the *upper limit topology* $\mathbb{R}_u$.